Chapter 5, Section 5

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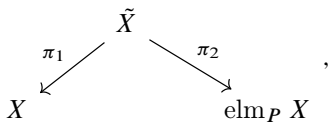
1. Let f be a rational function on the surface X. Show that it is possible to "resolve the singularities of f" in the following sense: there is a birational morphism $g : X' \to X$ so that f induces a morphism of X' to $\mathbb{P}^1$.

Proof. Assume f is nonconstant, and let $f : X \dashrightarrow \mathbb{P}^1$ be the rational map induced by f. By (I, Ex. 4.2), there is a largest open set $U \subset X$ on which f is represented by a morphism $\varphi : U \to \mathbb{P}^1$. If t is a point in $\mathbb{P}^1$, denote by $f^{-1}(t)$ the closure of $\varphi^{-1}(t)$ in X. The proof of (5.1) follows through for any rational map of smooth projective varieties. In particular, the locus of indeterminacy $Z = X - U$ of f is a finite set of points, and the fibers $\varphi^{-1}(t) \subset U$ are nonempty of dimension 1 (II, Ex. 3.22). Indeed, if $\varphi^{-1}(t)$ is empty for some t, then φ factors through $\mathbb{A}^1 \hookrightarrow \mathbb{P}^1$, which implies the existence of a regular function on U. This is not possible, because $\dim(X - U) = 0$, so every regular function on U extends uniquely to a regular function on X, and since X is projective, the only regular functions on X are the constant ones.

Now, we claim that if P is a closed point in X, then P is a point of indeterminacy of f if and only if P is contained in two distinct fibers of f. The reverse direction is clear by continuity, and conversely, if P is a point contained in only one fiber $f^{-1}(t)$, then the map $\varphi' : U \cup \{P\} \to \mathbb{P}^1$ defined by $\varphi'\big|_U = \varphi$ and $\varphi'(P) = t$ is also a morphism representing f, so by assumption on U being the largest open set for which f is a morphism, P must be in U. Write the divisor of f as $(f) = \sum n_i C_i$. Applying the embedded resolution (3.9) to the curve $Y = \bigcup C_i$ gives a birational morphism $\pi : \tilde{X} \to X$ such that $\pi^{-1}(Y)$ is a divisor with normal crossings. Let $\tilde{f} = f \circ \pi$, and let $\tilde{Z} \subset \tilde{X}$ be the locus of indeterminacy of $\tilde{f}$, which is also a finite set of points contained in $\pi^{-1}(Z)$. Similarly define the fiber $\tilde{f}^{-1}(t) = \pi^{-1}(f^{-1}(t))$. The divisor of $\tilde{f}$ can be written as a difference of effective divisors $(\tilde{f}) = E - D$, where $E = \sum a_i E_i$, $D = \sum b_j D_j$, every summand is an irreducible nonsingular curve, and whenever E_i, D_j intersect at a point $P \in \tilde{X}$, the local equations form a part of a regular system of parameters at P. The latter provision combined with (Ex. 3.2) implies that the strict transforms of E_i and D_j are disjoint after a sufficient number of blow ups of $\tilde{X}$ at P, and for the same reason as above, a point of intersection of E_i, D_j must be a point in $\tilde{Z}$. Let $\pi' : X' \to \tilde{X}$ be a composition of monoidal transformations so that $\pi'^{-1}(E)$ and $\pi'^{-1}(D)$ are disjoint, and let $f' = \tilde{f} \circ \pi'$. If $\tilde{E}$ and $\tilde{D}$ are disjoint, then $\tilde{E}.\tilde{D} = 0$, and since any two fibers of $\tilde{f}$ are linearly equivalent with a possible intersection at a point in $\tilde{Z}$, the strict transforms of every fiber $\tilde{f}^{-1}(t)$ in X' are also disjoint. Hence, f' is a morphism of X' to $\mathbb{P}^1$. □

5. Let C be a curve, and let $\pi : X \to C$ and $\pi' : X' \to C$ be two geometrically ruled surfaces over C. Show that there is a finite sequence of elementary transformations (5.7.1) which transform X into X'.

Proof. Let P be a point in X, and let L be the fiber of π containing P. Consider the elementary transform of X with center at P, which we denote by $\mathrm{elm}_P X$. Consider the diagram

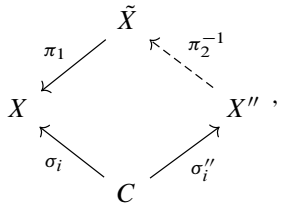


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where π_1 is the monoidal transformation of X centered at P with exceptional curve E, and if $\tilde{L}$ is the strict transform of L on $\tilde{X}$, then π_2 is the contraction of $\tilde{L}$, i.e., $Q = \pi_2(\tilde{L})$ is a point. Let $M = \pi_2(E)$, and let D be a section of π containing P. By (3.6), if $\tilde{D}$ is the strict transform of D on $\tilde{X}$, then $\tilde{D} \sim \pi_1^* D - E$, so if D' is the strict transform of D by elm_P, then $D' \sim \pi_{2*}\tilde{D}$. Also, π_2 is the monoidal transformation of $\mathrm{elm}_P X$ centered at Q, and D' does not meet Q, so $\pi_2^* D' = \tilde{D}$ by (3.6). Thus by (3.2), $D'^2 = \tilde{D}^2 = D^2 - 1$.

If X has invariant e, then it is defined by the projective bundle $\mathbb{P}(\mathcal{E})$, where $\mathcal{E}$ is a normalized locally free sheaf of rank 2, and $\bigwedge^2 \mathcal{E} \cong \mathcal{O}_C(\mathfrak{e})$ for some divisor $\mathfrak{e}$ of degree $-e$ (2.9). Let C_0 be the section of π corresponding to the surjection $\mathcal{E} \to \mathcal{O}_C(\mathfrak{e})$. If $D \sim C_0 + (\mathfrak{d} - \mathfrak{e})f$ is any section of π, then $\deg(\mathfrak{d} - \mathfrak{e}) \geq 0$ by the assumption that $\mathcal{E}$ is normalized. A consequence is $e \geq -D^2$, or expressed differently, $-e$ is the minimum value for the self-intersection of a section on a ruled surface. Let X_1 be an elementary transform of X centered at a point in C_0, and let $\tilde{C}_0$ be the strict transform of C_0. If X_1 has invariant e_1, then it follows by the above that $e_1 \geq -\tilde{C}_0^2 = e + 1$. Therefore, X can be transformed into a geometrically ruled surface with arbitrarily high invariant. By (2.12), $\mathcal{E}$ is decomposable if $e > 2g - 2$, so it remains to prove the statement for geometrically ruled surfaces determined by decomposable locally free sheaves. In fact, it suffices to show X can be transformed to the trivial ruled surface $\mathbb{P}(\mathcal{O}_C \oplus \mathcal{O}_C) \cong C \times \mathbb{P}^1$.

Now, let us consider how $\mathcal{E}$ behaves under elementary transformations of X if $\mathcal{E} = \mathcal{O}_C \oplus \mathcal{O}_C(\mathfrak{e})$ and $e = -\deg \mathfrak{e} > 0$. Ruled surfaces determined by decomposable locally free sheaves have two distinguished sections C_0, C_1 corresponding to the surjections of $\mathcal{E}$ onto each summand $\mathcal{O}_C(\mathfrak{e}), \mathcal{O}_C$, respectively, such that $C_0^2 = -e$, $C_1^2 = e$, and $C_0 \cap C_1 = \emptyset$. Let P be a point in X lying over $R = \pi(P) \in C$, and consider the elementary transform X'' of X centered at P. Following the notation of the above, we have a commutative diagram



where σ_i, σ_i'' are sections defining either $C_i, \tilde{C}_i$ for each $i = 0, 1$. Let $\tilde{C}_0, \tilde{C}_1$ denote the strict transforms of C_0, C_1, respectively. There are two cases we will consider, which are $P \in C_0$ or $P \in C_1$, and in either case, $\tilde{C}_0 \cap \tilde{C}_1 = \emptyset$, so X'' is decomposable by (Ex. 2.2).

$\boxed{P \in C_0}$ We claim $X'' \cong \mathbb{P}(\mathcal{O}_C \oplus \mathcal{O}_C(\mathfrak{e} - R))$. Under the identifications $\tilde{C}_i \cong C$ for each $i = 0, 1$ via σ_i'', we have the isomorphisms

$$\mathcal{O}_{X''}(\tilde{C}_0) \otimes \mathcal{O}_{\tilde{C}_0} \cong \mathcal{O}_C(\mathfrak{e} - R), \quad \mathcal{O}_{X''}(\tilde{C}_1) \otimes \mathcal{O}_{\tilde{C}_1} \cong \mathcal{O}_C.$$

Indeed, $\pi_2^{-1} \circ \sigma_0''$ is a morphism, and π_2^{-1} is an isomorphism outside of the strict transform of the fiber containing P, so there is an isomorphism of invertible sheaves under the identification $\tilde{C}_0 \cong C$,

$$\mathcal{O}_{X''}(\tilde{C}_0) \otimes \mathcal{O}_{\tilde{C}_0} \cong \sigma_0''^* \mathcal{O}_{X''}(\tilde{C}_0) \cong (\pi_2^{-1} \circ \sigma_0'')^* \mathcal{O}_{\tilde{X}}(\pi_1^* C_0 - E) \cong \mathcal{O}_C(\mathfrak{e} - R),$$

and similarly for $\mathcal{O}_{X''}(\tilde{C}_1) \otimes \mathcal{O}_{\tilde{C}_1} \cong \mathcal{O}_C$. We conclude by the proof of (Ex. 2.2) that $X'' \cong \mathbb{P}(\mathcal{E}'')$, where $\mathcal{E}'' = \mathcal{O}_C \oplus \mathcal{O}_C(\mathfrak{e} - R)$.

$\boxed{P \in C_1}$ We claim $X'' \cong \mathbb{P}(\mathcal{O}_C \oplus \mathcal{O}_C(\mathfrak{e} + R))$. Similar to the previous case, we have the isomorphisms

$$\mathcal{O}_{X''}(\tilde{C}_0) \otimes \mathcal{O}_{\tilde{C}_0} \cong \mathcal{O}_C(\mathfrak{e}), \quad \mathcal{O}_{X''}(\tilde{C}_1) \otimes \mathcal{O}_{\tilde{C}_1} \cong \mathcal{O}_C(-R).$$

Therefore, $X'' \cong \mathbb{P}(\mathcal{E}'')$, where $\mathcal{E}'' = \mathcal{O}_C(-R) \oplus \mathcal{O}_C(\mathfrak{e})$, which is isomorphic to the projective bundle of $\mathcal{O}_C \oplus \mathcal{O}_C(\mathfrak{e} + R)$.

If we write $\mathfrak{e} = \sum n_i P_i$, performing the appropriate elementary transforms centered at points in the fibers over P_i will give a ruled surface $X'' = \mathbb{P}(\mathcal{E}'')$ with $\mathcal{E}'' = \mathcal{O}_C \oplus \mathcal{O}_C$. Hence, any ruled surface over C can be transformed to $C \times \mathbb{P}^1$ with a finite sequence of elementary transforms. □